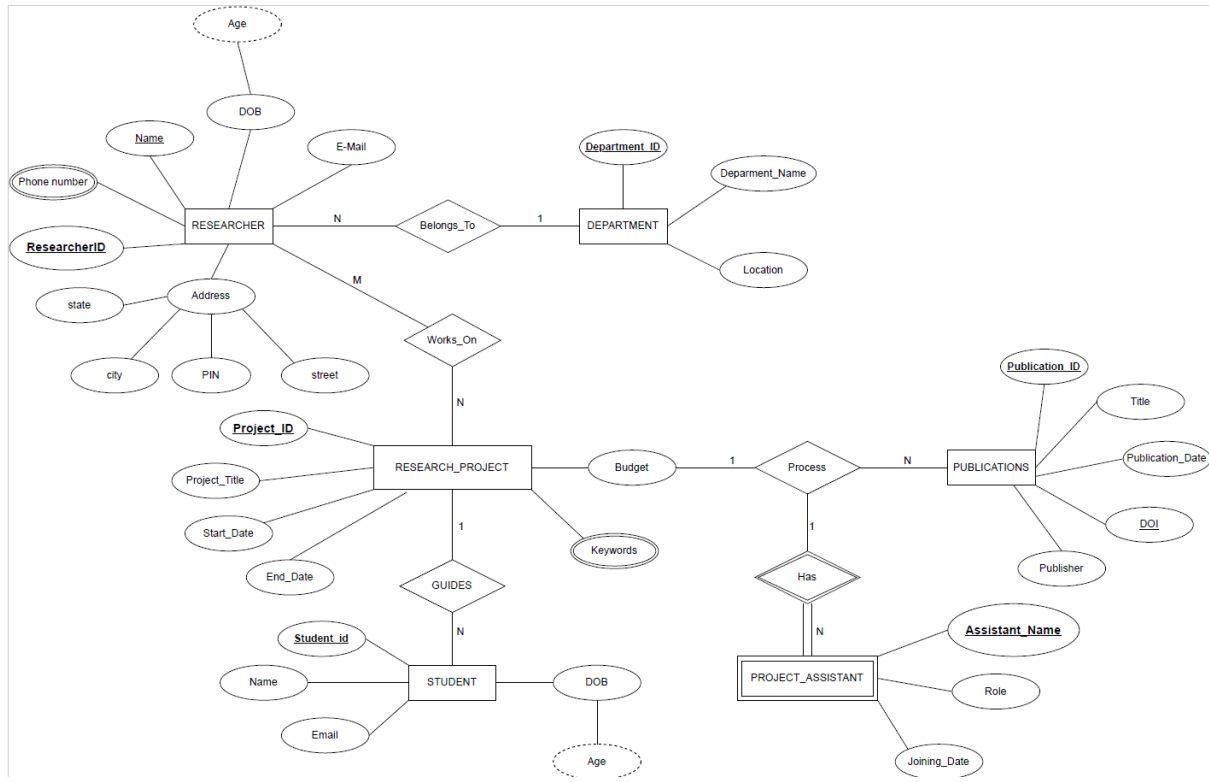


LAB-3:**ER-Diagram:****1. Identify all entity types and entity sets**

- Researcher: Represents all research staff members.
- Department: Represents the administrative departments.
- Research Project: Represents individual research initiatives.
- Student: Represents students working under researchers.
- Publication: Represents research outputs produced from projects.
- Project Assistant: Represents assistants hired within the scope of a project.

2. Identify strong and weak entity sets

- Strong Entities: Researcher, Department, Research Project, Student, Publication
- Weak Entity: Project Assistant (depends on Research Project, no independent primary key)

3. Identify the owner entity and partial key of the weak entity

- Owner Entity: Research Project
- Partial Key (Discriminator): Assistant_Name

4. Identify simple attributes

- Researcher: Researcher_ID, Name, Date_of_Birth, Email
- Address Components: Street, City, State, PIN Code
- Department: Department_ID, Department_Name, Location
- Research Project: Project_ID, Project_Title, Start_Date, End_Date, Budget
- Student: Student_ID, Name, Email, Date_of_Birth
- Publication: Publication_ID, Title, Publication_Date, DOI, Publisher
- Project Assistant: Assistant_Name, Role, Joining_Date

5. Identify composite attributes

- Address: Street, City, State, PIN Code

6. Identify single-valued attributes

- All attributes except Phone Numbers and Keywords

7. Identify multi-valued attributes

- Phone Numbers (Researcher)
- Keywords (Research Project)

8. Identify stored attributes

- Stored: Date_of_Birth (Researcher & Student), Email, Address, Budget, Publisher, etc.

9. Identify derived attributes

- Derived: Age (calculated from Date_of_Birth)

10. Identify key attributes

- Researcher: Candidate Keys → Researcher_ID, Email; Primary Key → Researcher_ID
- Department: Candidate Keys → Department_ID, Department_Name; Primary Key → Department_ID
- Research Project: Candidate Key → Project_ID; Primary Key → Project_ID
- Student: Candidate Keys → Student_ID, Email; Primary Key → Student_ID
- Publication: Candidate Keys → Publication_ID, DOI; Primary Key → Publication_ID
- Project Assistant: Composite Key → (Project_ID, Assistant_Name)

11. Identify candidate keys and select appropriate primary keys

- Researcher: PK → Researcher_ID
- Department: PK → Department_ID
- Research Project: PK → Project_ID
- Student: PK → Student_ID
- Publication: PK → Publication_ID
- Project Assistant: PK → (Project_ID, Assistant_Name)

12. Identify at least one composite key

- Project Assistant: (Project_ID, Assistant_Name)
- Works_On (Bridge Table): (Researcher_ID, Project_ID)

13. Identify appropriate foreign keys after converting the ER model into relations

- Researcher Table → Department_ID (FK)
- Student Table → Researcher_ID (FK)
- Publication Table → Project_ID (FK)
- Project_Assistant Table → Project_ID (FK, part of composite PK)
- Works_On Table → Researcher_ID (FK), Project_ID (FK), composite PK (Researcher_ID, Project_ID)
- Researcher_Phone Table → Researcher_ID (FK)

- Project_Keywords Table → Project_ID (FK)

14. Identify all major relationships between the entities

- Researcher — Belongs_To (N:1) — Department
- Researcher — Works_On (M:N) — Research Project
- Researcher — Supervises (1:N) — Student
- Research Project — Produces (1:N) — Publication
- Research Project — Employs (1:N) — Project Assistant (Identifying relationship for weak entity)

15. Draw the complete ER diagram using standard ER notation

- Use rectangles for strong entities, double rectangles for weak entities.
- Diamonds for relationships, double diamonds for identifying relationships.
- Ovals for attributes, double ovals for multi-valued, dashed ovals for derived.
- Connect with proper cardinalities (N:1, M:N, 1:N).

16. Prepare a table showing each attribute and its attribute type and key status

Entity/Attribute	Type	Key Status
Researcher_ID	Simple	Primary Key
Name	Simple	-
Date_of_Birth	Simple	Stored
Age	Derived	-
Email	Simple	Candidate Key
Phone Numbers	Multi-Valued	-
Address (Street, City, State, PIN)	Composite	-
Department_ID	Simple	Primary Key
Department_Name	Simple	Candidate Key
Location	Simple	-

Entity/Attribute	Type	Key Status
Project_ID	Simple	Primary Key
Project_Title	Simple	-
Start_Date	Simple	-
End_Date	Simple	-
Budget	Simple	-
Keywords	Multi-Valued	-
Student_ID	Simple	Primary Key
Student_Name	Simple	-
Student_Email	Simple	Candidate Key
Student_DOB	Simple	Stored
Publication_ID	Simple	Primary Key
Title	Simple	-
Publication_Date	Simple	-
DOI	Simple	Candidate Key
Publisher	Simple	-
Assistant_Name	Simple	Partial Key
Role	Simple	-
Joining_Date	Simple	-